NBAyes: Bayesian Modeling of NBA Athlete's Career Trajectories

Abhijit Brahme & Alex Franks

Overview

- Discuss a novel prior on concave functions to encode realistic aging dynamics.

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- Illustrate application of this prior in a Bayesian model of NBA player trajectories.
- Discuss inference challenges and present preliminary results on trajectory estimation and injury impact analysis.

Concavity

A function $f(t)$ is concave on some interval $[a, b]$ if it's second derivative is non-positive:

$f''(t) \leq 0$ for all $t \in [a, b]$.

Therefore, any function of the form

$$f(t) = \beta_0 + \beta_1(t - t_0) + \int_{t_0}^{t} \int_{t_0}^{s} f''(z)\, dz\, ds,$$

is concave as long as $f''(t) \leq 0$ for all t.

Theory: Prior on Concave Functions

Proposition (Concave function prior). Define

$$h(t) = \beta_0 + \beta_1(t - t_0) - \int_{t_0}^{t} \int_{t_0}^{s} [g(z)]^2 \, dz \, ds,$$

with $g(z)$ a Gaussian process; this induces a concave prior over continuous trajectories.

Theory: Prior on Concave Functions

Proof sketch.

The prior over h is induced by the Gaussian process prior over g, which is a valid probability measure over continuous functions. Hence, this *continuous* transformation $T(g)$ defines a valid prior over concave functions.

$$T(g) = \beta_0 + \beta_1(t - t_0) - \int_{t_0}^{t} \int_{t_0}^{s} g(z)^2 \, dz \, ds$$

Theory: Peak Constraints

The following constraints on β_0 and β_1 ensure a peak of f at $t_{\max}$ with value $f_{\max}$:

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$$\beta_1 = \int_{t_0}^{t} [g(z)]^2 \, dz$$

$$\beta_0 = f_{\max} - \beta_1 (t_{\max} - t_0) + \int_{t_0}^{t_{\max}} \int_{t_0}^{s} [g(z)]^2 \, dz \, ds$$

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$$f(t) = f_{\max} + (t - t_{\max}) \int_{t_0}^{t_{\max}} g(z)^2 \, dz - \int_{t}^{t_{\max}} \int_{t_0}^{s} g(z)^2 \, dz \, ds$$

Theory: Behavior of Concave Prior

Expected Value of f(t) (conditioning on $t_{\max}, f_{\max}$):

- Start from the constrained concave prior

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- Take conditional expectation term-by-term:

$$\begin{aligned} E[f(t) \mid t_{\max}, f_{\max}] = f_{\max} & \\ & + (t - t_{\max}) \int_{t_0}^{t_{\max}} E[g(z)^2] dz \\ & - \int_{t}^{t_{\max}} \int_{t_0}^{s} E[g(z)^2] dz \, ds. \end{aligned}$$

Theory: Behavior of Concave Prior (cont.)

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- Substitute and integrate:

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\begin{aligned}
E[f(t) \mid t_{\max}, f_{\max}] = f_{\max} & \\
& + \sigma^2 (t_{\max} - t_0)(t - t_{\max}) \\
& - \sigma^2 \frac{(t_{\max} - t)^2}{2}.
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- Simplify:

$$E[f(t) \mid t_{\max}, f_{\max}] = -\frac{\sigma^2}{2}(t - t_0)^2 + \frac{\sigma^2}{2}(t_{max} - t_0)^2 + f_{\max}$$

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- This is intuitive, as it is a parabolic function with maximum at $t_{\max}$, value $f_{\max}$, and curvature determined by σ^2.

Theory: Behavior of Concave Prior (cont.)

- The variance of $f(t)$ is determined by the variance of the double integral of the squared GP

Variance of f(t) (conditioning on $t_{\max}, f_{\max}$ fixed):

$$f(t) = f_{\max} + (t - t_{\max}) \int_{t_0}^{t_{\max}} g(z)^2 \, dz - \int_{t}^{t_{\max}} \int_{t_0}^{s} g(z)^2 \, dz \, ds.$$

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- take definition of variance:

$$Var[f(t) \mid t_{\max}, f_{\max}] = E[f(t)^2 \mid t_{\max}, f_{\max}] - E[f(t) \mid t_{\max}, f_{\max}]^2$$

Theory: Behavior of Concave Prior (cont.)

Roughly (omitting a lot of math), the variance grows quartically as we move away from the peak, with curvature determined by σ^4:

$$Var[f(t) \mid t_{\max}, f_{\max}] \propto 2\sigma^4 (t - t_{max})^4$$

- Why (Isserlis Theorem)?

$$\begin{aligned}
E[f(t)^2] &\approx E\left[\left(\int_t^{t_{max}} \int_t^s g(z)^2 dz ds\right)^2\right] \\
&\approx E\left[\int_t^{t_{max}} \int_t^s \int_t^{t_{max}} \int_t^{s'} g(z)^2 g(z')^2 dz dz' ds ds'\right] \\
&\approx \int_t^{t_{max}} \int_t^s \int_t^{t_{max}} \int_t^{s'} E[g(z)^2 g(z')^2] dz dz' ds ds' \\
&\approx 3\sigma^4 \frac{(t - t_{max})^4}{4}
\end{aligned}$$

Theory: Practically Squaring a GP

1. HSGP approximation (Solin and Särkkä, 2020):

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3. Squared exponential spectral density:

$$S(\omega) = \sigma^2 \sqrt{2\pi}\, l \, \exp\left(-\frac{1}{2} l^2 \omega^2\right) = \int_{-\infty}^{\infty} \exp\left(-\frac{\tau^2}{2l^2}\right) e^{-i\omega\tau} d\tau$$

Theory: Practically Squaring a GP (cont.)

Let

$$\tilde{\alpha}_l = S(\sqrt{\lambda_l})^{1/2}\alpha_l, c_{lm} = \tilde{\alpha}_l\tilde{\alpha}_m.$$

Then the squared process can be written compactly as:

$$[g(t)]^2 \approx \left(\sum_{l=1}^{K} \tilde{\alpha}_l \psi_l(t)\right)^2 = \sum_{l=1}^{K}\sum_{m=1}^{K} c_{lm}\, \psi_l(t)\psi_m(t).$$

Hence,

$$\int_{t_0}^{t}\int_{t_0}^{s} [g(z)]^2\, dz\, ds \approx \sum_{l=1}^{K}\sum_{m=1}^{K} c_{lm} \int_{t_0}^{t}\int_{t_0}^{s} \psi_l(z)\psi_m(z)\, dz\, ds.$$

Theory: Analytical Solution for Double Integral

Recall that $\psi_l(z) = \sqrt{\frac{1}{L}}\sin(\sqrt{\lambda_l}(L+z))$:

$$\int_{t_0}^{t}\int_{t_0}^{s}\psi_l(z)\psi_m(z)\,dz\,ds = \Psi_{lz}(t),$$

$$\Psi_{lz}(t) = \begin{cases} \dfrac{\cos(\epsilon_{lz}^{+}(t+L))-1}{2L(\epsilon_{lz}^{+})^2} + \dfrac{(t+L)^2}{4L}, l = z, \\ \dfrac{\cos(\epsilon_{lz}^{+}(t+L))-1}{2L(\epsilon_{lz}^{+})^2} - \dfrac{\cos(\epsilon_{lz}^{-}(t+L))-1}{2L(\epsilon_{lz}^{-})^2}, l \neq z. \end{cases}$$

$\epsilon_{lz}^{+} = \sqrt{\lambda_l} + \sqrt{\lambda_z}$, $\epsilon_{lz}^{-} = \sqrt{\lambda_l} - \sqrt{\lambda_z}$, $t_0 = -L$

Theory: HSGP Approx. Convergence

Let k be a stationary covariance function on $[0, T]$ with spectral density S, and let $\{\phi_i, \lambda_i\}_{i=1}^{\infty}$ be the Laplacian eigenpairs on $[0, T]$. Define $\omega_i = \sqrt{\lambda_i}$. Let

$$g(t) = \sum_{i=1}^{\infty} \sqrt{S(\omega_i)}\, \alpha_i \psi_i(t), \qquad \alpha_i \overset{iid}{\sim} N(0, 1),$$

and its truncation

$$g_K(t) = \sum_{i=1}^{K} \sqrt{S(\omega_i)}\, \alpha_i \psi_i(t).$$

Define

$$f(t) = \int_0^t \int_0^s g(z)^2 \, dz \, ds, \qquad f_K(t) = \int_0^t \int_0^s g_K(z)^2 \, dz \, ds.$$

Then

$$f_K \overset{D}{\to} f \quad [0, T].$$

Theory: HSGP Approx. Convergence (cont.)

Proof sketch. By the HSGP construction, the covariance operator admits the spectral approximation

$$k(t, t') \approx \sum_{i=1}^{K} S(\omega_i)\psi_i(t)\psi_i(t').$$

which approximates the true covariance of g uniformly as $K \to \infty$ (Solin and Sarkkä, 2020).

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- the sum g_k is mean-zero Gaussian with covariance function given by the spectral approximation above
- the finite-dimensional distributions of g_K converge to those of g, since the covariances coincide in the limit.
- $I(g) = \int_0^t \int_0^s g(z)^2 \, dz \, ds$ is continuous, the finite-dimensional distributions of $f_K = I(g_K^2)$ converge to those of $f = I(g^2)$ in distribution.

Prior Predictive Behavior

Show demo

Adding Additional Dimensions

Let's say we have additional covariates X that we want to condition on. We can extend the concave prior construction as follows to maintain concavity in t for each fixed X:

$$f(X,t) = \beta_0 + \beta_1(t - t_0) - \int_{t_0}^{t} \int_{t_0}^{s} [g(z,X)]^2 \, dz \, ds,$$

where $g(z,X)$ is a Gaussian process over the joint space of z and X. In our case, $g(z,X) \sim GP(0, k(z,z')k(X,X'))$, where the covariance function is separable over the time and covariate spaces.

Adding Additional Dimensions (cont.)

The same constraints can be applied to ensure a peak at $t_{\max}(X)$ for each fixed X, with value $f_{\max}(X)$:

1. $f(X, t_{\max}(X)) = f_{\max}(X)$

Adding Additional Dimensions (cont.)

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1. $f(X, t_{\max}(X)) = f_{\max}(X)$
2. $\left.\frac{d}{dt} f(X,t)\right|_{t=t_{\max}(X)} = 0$

Adjusting the HSGP Approx.

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- Squaring/integration steps are unchanged; only coefficients now vary with X.

Adjusting the HSGP Approx. (cont.)

Is the covariance of $g(z, X)$ still approximately separable in z and X? **Proof sketch:**

$$\mathrm{Cov}(g(z, X), g(z', X')) = \mathrm{Cov}\left(\sum_{l=1}^{K} S(\sqrt{\lambda_l})\psi_l(z)\alpha_l(X), \sum_{m=1}^{K} S(\sqrt{\lambda_m})\psi_m(z')\alpha_m(X')\right)$$

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&= \sum_{l=1}^{K}\sum_{m=1}^{K} S(\sqrt{\lambda_l})^{1/2} S(\sqrt{\lambda_m})^{1/2}\psi_l(z)\psi_m(z') \\
&\qquad \cdot\ \mathrm{Cov}(\alpha_l(X), \alpha_m(X'))
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&= \sum_{l=1}^{K} S(\sqrt{\lambda_l})\psi_l(z)\psi_l(z')\,\mathrm{Cov}(\alpha_l(X), \alpha_l(X')) \\
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&= k(X,X')\sum_{l=1}^{K} S(\sqrt{\lambda_l})\psi_l(z)\psi_l(z') \\
&\approx k(X,X')k(z,z')
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Adjusting the HSGP Approx. (cont.)

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- Then $\alpha_l(X) = w_l^T X$ (Bayesian factor model form).
- Each basis function has its own weight vector $w_l \sim N(0, I)$.

Adjusting the HSGP Approx. (cont.)

Some Notes:

- Different kernels $k(X, X')$ can be used to capture different covariate effects on trajectory shape.

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- Framework remains valid as long as $\alpha_l(X)$ are GP-distributed.
- Can apply HSGP/linearized approximations over $k(X, X')$ for scalability.
- resulting process $f(X, t)$ is not separable due to transformations

Concave Prior for Non-Gaussian Response

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$$Y(X, t) \sim \text{Poisson}(\exp(\eta(X, t)))$$

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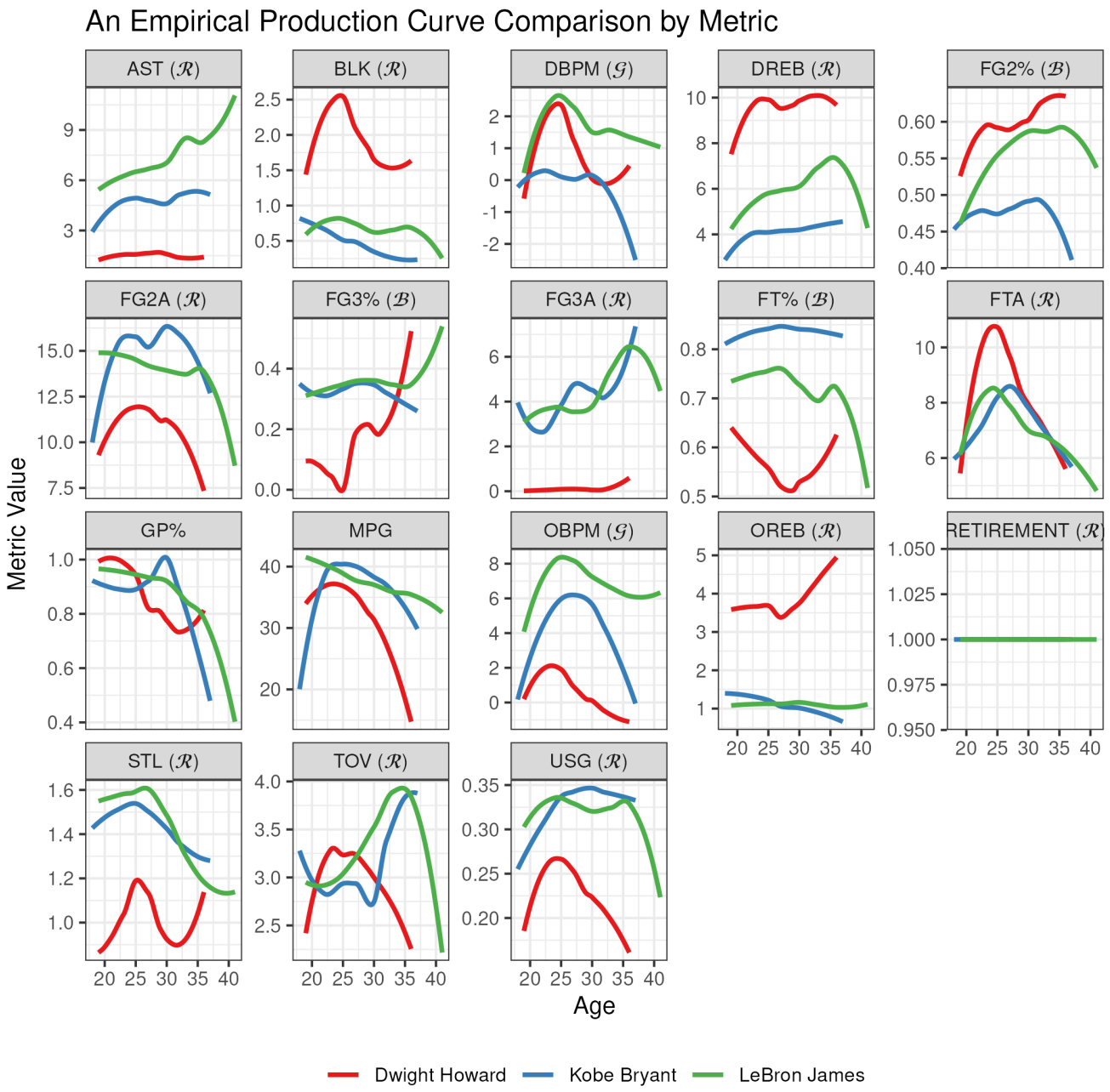
$$Y(X, t) \sim \text{Poisson}(\exp(\eta(X, t)))$$

- Since the exponential link is monotone, peak location and peak value priors can still be effective, while concavity isn't necessarily preserved in the observed data space

Why a Concave Prior?



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- "P" $\approx$ 2800 players with at least 1 season of data.
- "T" = 20 seasons of potential career length (age 18-38).

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- Right censored data for players who are still active or whose careers end after 38.
- Left censored data for players who start after age 18 or whose careers start before 1997.
- Mix of informative and uninformative missingness patterns across metrics and seasons.

Model Assumptions

1. Player heterogeneity is captured by latent embeddings.

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$$f^k(X, t) \sim MCP(\Theta; X)$$

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3. Injuries induce a shock and recovery trajectory that is learned from data.

$$\tau_{ptm}^{k} \sim N(w_m^T \tau_k^*, \sigma_k^2)$$

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1. Player heterogeneity is captured by latent embeddings.
2. Player trajectories are smooth and concave in age, with a single peak.
3. Injuries induce a shock and recovery trajectory that is learned from data.
4. Short-term deviations around the aging curve are modeled as a noise process.

$$r_p^k(t) \sim AR(1)$$

Model Assumptions

1. Player heterogeneity is captured by latent embeddings.
2. Player trajectories are smooth and concave in age, with a single peak.
3. Injuries induce a shock and recovery trajectory that is learned from data.
4. Short-term deviations around the aging curve are modeled as a noise process.
5. Observational noise is heteroscedastic and metric-specific.

$$
\begin{aligned}
\mu_{ptm}^{k} &= f^{k}(X_p, t) + \tau_{ptm}^{k} + r_p^{k}(t) \\
\eta_{pt}^{k} &= \mu_{pt}^{k} + \mathbb{1}_{\{k \in \mathcal{K}_{\text{rate}}\}} \log e_{pt}^{k} \\
Y_{pt}^{k} &\sim \mathrm{EF}\big(\eta_{pt}^{k},\, \phi_{pt}^{k}\big)
\end{aligned}
$$

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- Bayesian functional model for joint skill + athleticism trajectories.

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- Handles heteroscedastic uncertainty, concave aging structure, and latent player embeddings.
- Explicitly estimates injury impact and counterfactual recovery paths.
- Metric correlation is not explicitly modeled, but shared latent structure induce correlation in practice.

Model Inference Challenges

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- Identifiability issues between player embeddings, injury effects, and aging curves
- A rotation matrix R can be applied to the latent space such that $\alpha_l(X) = w_l^T R R^{-1} X$, which preserves the likelihood but creates non-identifiability between w_l and X.

Proposed Inference Approach

Addressing Additive Components and Identifiability

1. Fit the concave function and AR components first to the data (mask out injured time-points from the likelihood) without the injury component to get a MAP estimate of the aging curve and AR parameters.

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1. Fit the concave function and AR components first to the data (mask out injured time-points from the likelihood) without the injury component to get a MAP estimate of the aging curve and AR parameters.
2. Fix the concave function and AR parameters to their MAP estimates, and then fit the injury component to the residuals to get a MAP estimate of injury effects.

Proposed Inference Approach (cont.)

Addressing Rotational Ambiguity in Latent Space

1. Initialize parameter values using the MAP procedure found by Stochastic Gradient Descent (SGD).

Proposed Inference Approach (cont.)

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Proposed Inference Approach (cont.)

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3. To address rotational ambiguity for the latent factor models, fix factor loadings to their MAP estimates, and allow the latent embeddings to vary during MCMC sampling. (Mauri and Dunson, 2025)
4. Run MCMC sampling (e.g. NUTS) over the remaining parameters to obtain posterior samples for trajectory estimation and injury impact analysis.

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- AR(1) noise process with metric-specific parameters, ρ^k and σ^k.
- Latent player embeddings of dimension $D = 21$.
- $f_{max}(X), t_{max}(X)$ modeled as linear functions of the latent embeddings centered around empirical means.

Model Results

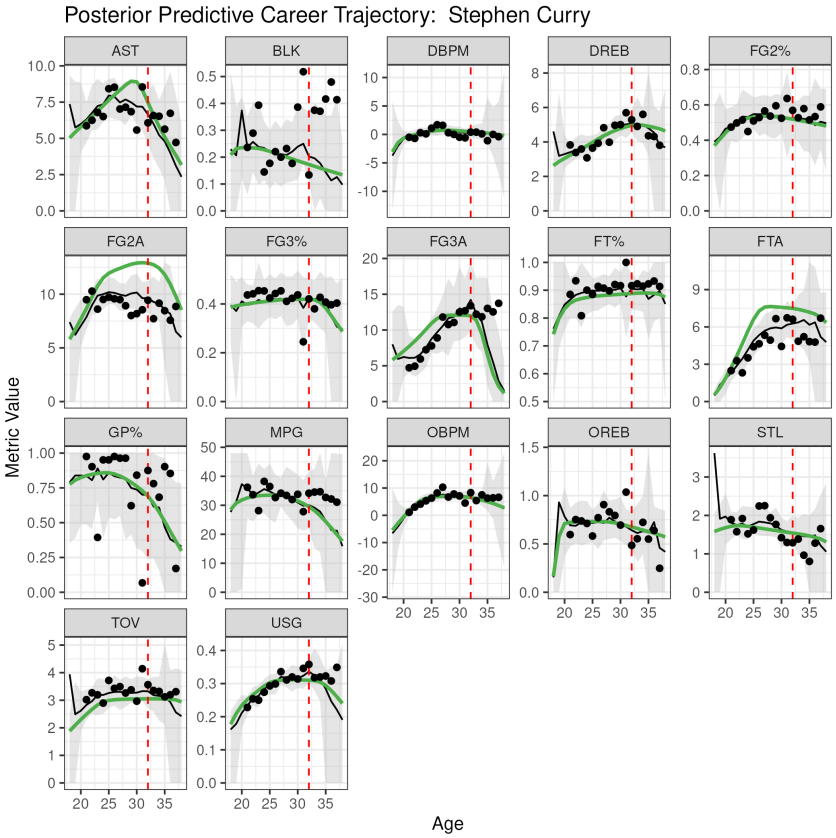


Figure 1: Stephen Curry Forecasted Trajectory

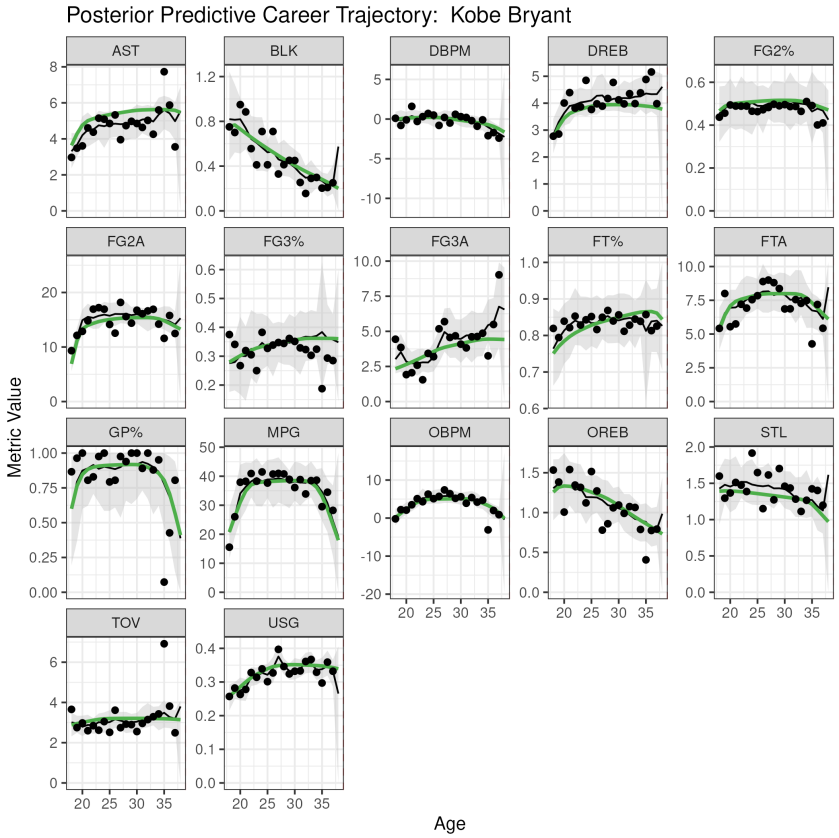
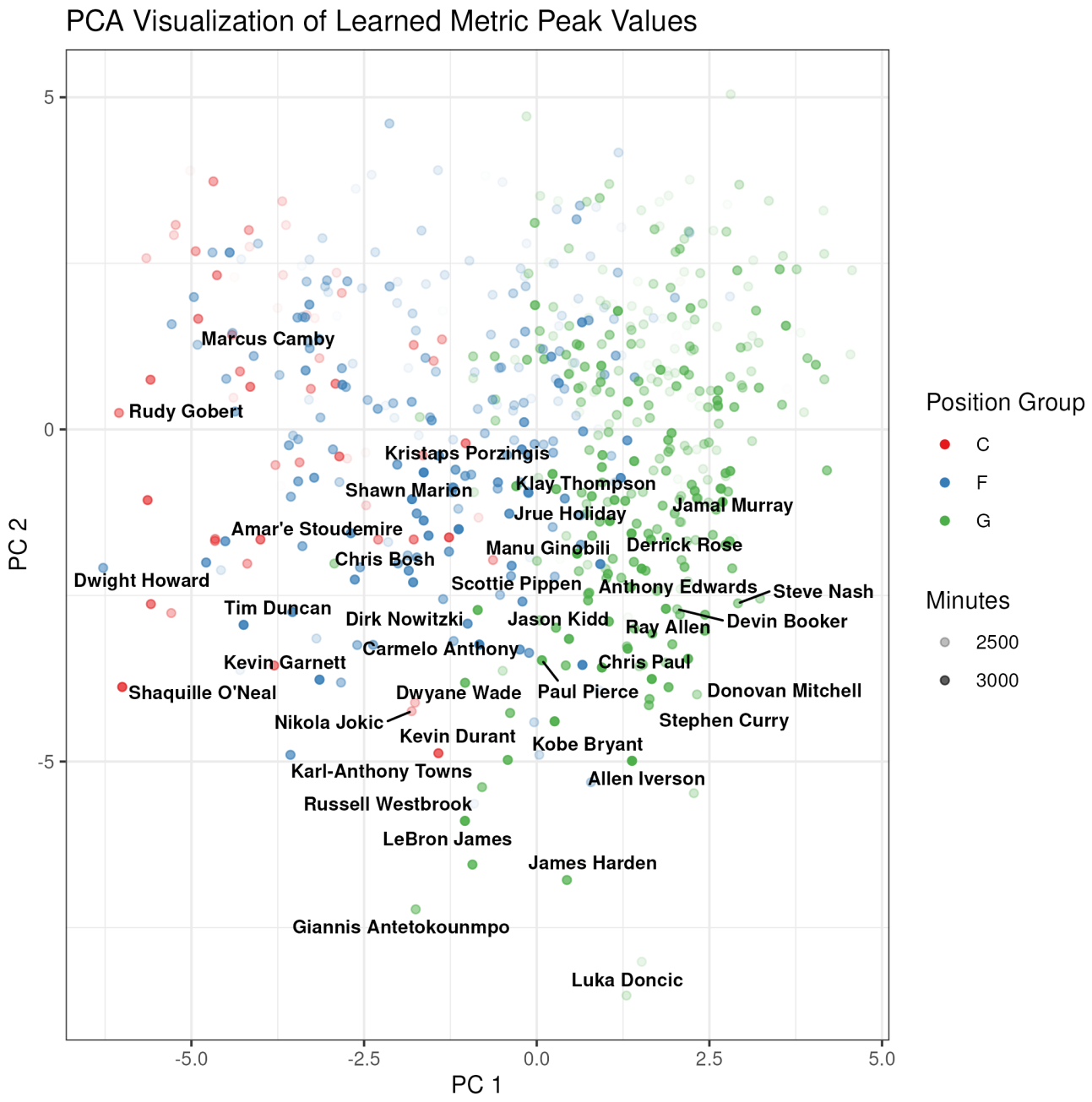
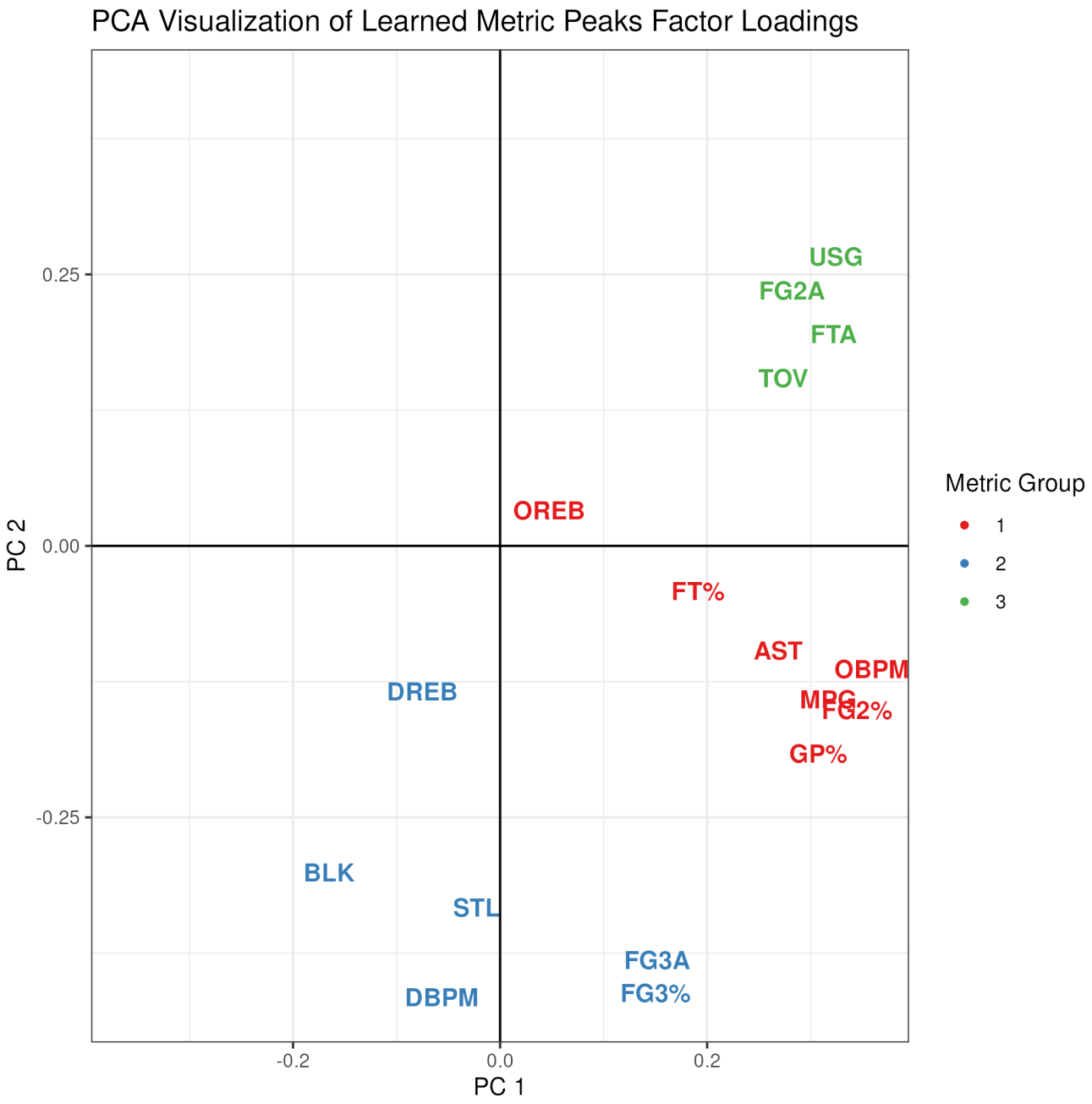


Figure 2: Kobe Bryant Forecasted Trajectory

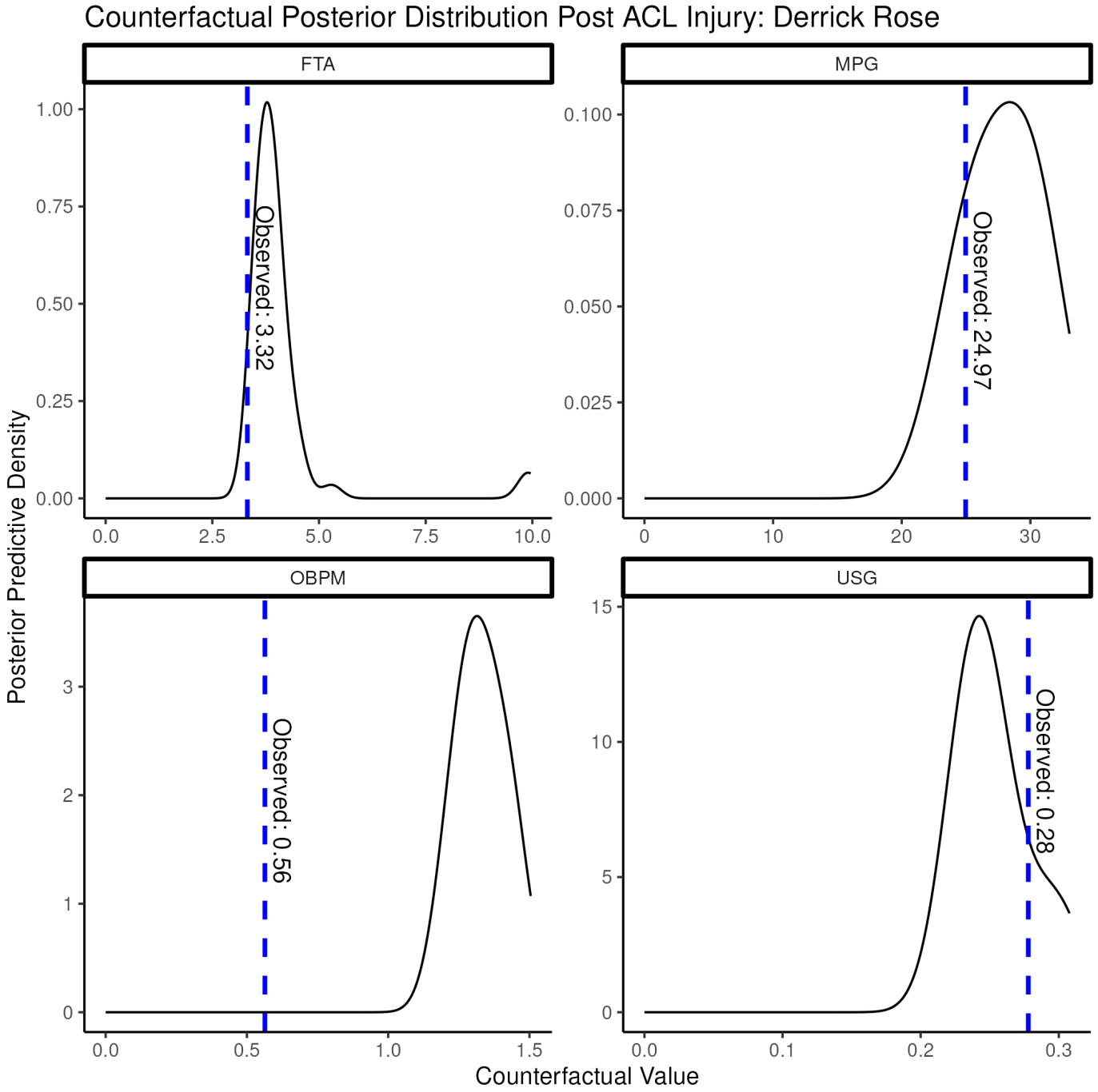
Model Results: PCA



Model Results: Metric Types



Injury Impact Estimation



Injury Impact Estimation: Population Posterior

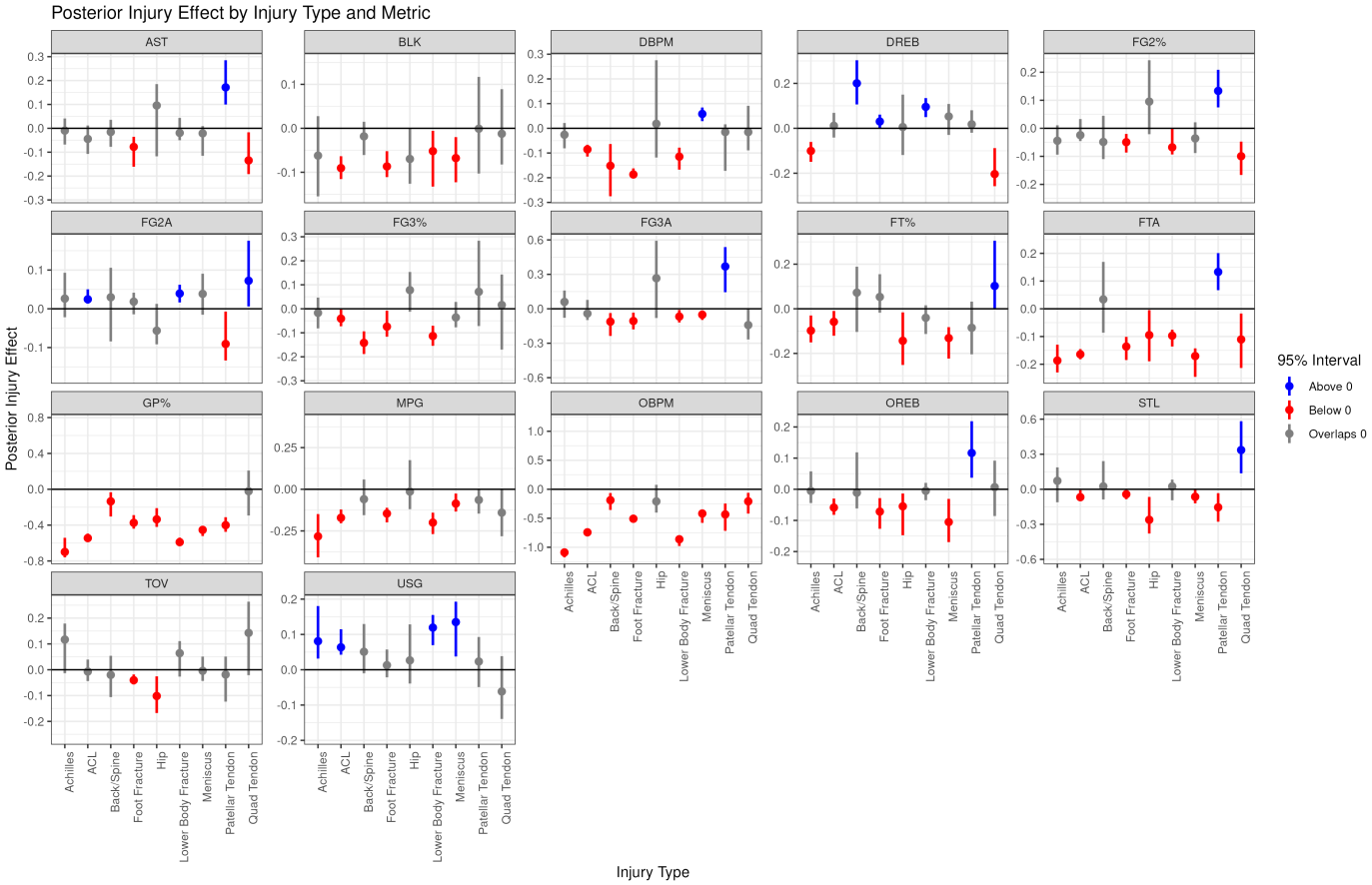


Figure 4: Posterior interval of injury mean effect by metric.

Future Work

1. Model Validation

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3. Injury Heterogeneity Analysis